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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 6.4

The Gram-Schmidt Process

The Gram-Schmidt Process

Goal: Create an orthogonal (or orthonormal) basis for any nonzero subspace of $\mathbb{R}^n$.

Example 1

Let $W = \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$. Construct an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for W . Then, construct an orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2\}$.

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix}$$

The Gram-Schmidt Process

Given a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned}\mathbf{v}_1 &= \mathbf{x}_1 \\ \mathbf{v}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 \\ \mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 \\ &\vdots\end{aligned}$$

Then, $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is an orthogonal basis for W .

The Gram-Schmidt Process

Furthermore, note that

$$\text{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$$

for $1 \leq k \leq p$.

Also note that we can scale any of the vectors $\mathbf{v}_i$ by a nonzero scalar without affecting the orthogonality of $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$.

Example 2

Find an orthogonal basis for the column space of

$$A = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 1 & 1 \\ -1 & 5 & -2 \\ 3 & -7 & 8 \end{bmatrix}$$

The QR Factorization

If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as

$$A = QR$$

where Q is an $m \times n$ matrix whose columns form an orthonormal basis for the column space of A and R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

Example 3

Find a QR factorization of

$$A = \begin{bmatrix} 5 & 9 \\ 1 & 7 \\ -3 & -5 \\ 1 & 5 \end{bmatrix}$$

Example 4

Find a QR factorization of

$$A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}$$
